

Predictive Modelling of IPO Listing Gains

Domain: Capital Markets | Tools: SPSS, Excel | Data: 258 Mainboard IPOs, NSE/BSE (2016–2025)

Summary

Objective

Test which pre-listing IPO signal — Grey Market Premium (GMP), subscription demand, or media sentiment — actually predicts listing-day gains, so investors can move from hype-driven decisions to evidence-based ones.

Concepts / Methods Used

Multiple Linear Regression (3 model specifications), Pearson Correlation, Collinearity Diagnostics (Tolerance/VIF), Log Transformation (skew correction), Z-score Outlier Trimming with re-validation.

Key Output

- Full model: $R^2 = 0.638$ ($F = 88.93$, $p < .001$) — combined signals explain 64% of listing-gain variation
- GMP is the dominant predictor ($\beta = .705$, $p < .001$); subscription and sentiment lose significance in its presence
- Without GMP, subscription demand alone is significant ($R^2 = .419$, all three categories $p < .05$)
- Media sentiment is insignificant in every model tested ($p = .25$ to $.59$)
- Finding holds after outlier removal (258 → 239 IPOs; GMP remains significant at $p < .001$)

Business Actionables

- Use GMP as the primary screening signal for listing-gain potential
- Use QIB/HNI subscription as a secondary confirmation check — a mismatch (high GMP with weak QIB demand) is a warning flag
- Discount media buzz and hype — it has no measured predictive value
- Evaluate on the full outcome distribution (range: -82% to $+253\%$), not the average gain, as downside risk is material

Detailed Analysis — Actual SPSS Output, Explained, With Actionables

Output 1 — Descriptive Statistics (N = 258)

Descriptive Statistics						
	N	Minimum	Maximum	Mean	Std. Deviation	Skewness
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic
Log_retail	258	-1.3979	2.2744	.732218	.6374631	.085
Log_HNI	258	-1.6990	2.9674	1.051653	.9892618	-.097
Log_QIB	258	-.7696	2.4494	1.277516	.7665445	-.339
Listing_gain	258	-82.00	252.76	23.6486	40.04584	1.947
Net_sent	258	-4	10	.21	1.618	1.135
GMP_percentage	258	-8.7678	99.2857	21.892812	24.7428103	1.256
Valid N (listwise)	258					

Descriptive Statistics			
	Skewness	Kurtosis	
	Std. Error	Statistic	Std. Error
Log_retail	.152	-.271	.302
Log_HNI	.152	-.885	.302
Log_QIB	.152	-1.024	.302
Listing_gain	.152	6.004	.302
Net_sent	.152	7.071	.302
GMP_percentage	.152	.611	.302
Valid N (listwise)			

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Regression

Actual SPSS output: descriptive statistics for all key variables

What this is: SPSS summary of every key variable — mean, minimum, maximum, standard deviation, skewness, and kurtosis — computed on all 258 IPOs before any modelling was done.

Layman meaning: The average listing gain across all IPOs was about 23.6%, but the standard deviation (40.05) is nearly double the mean, and gains ranged from -82% (a large loss) to +252.76% (an extreme gain). Listing-day outcomes are extremely inconsistent — a headline "average IPO gains 24%" statistic hides serious risk. GMP and subscription variables also show heavy skew, which is why they needed transformation before modelling.

Actionable for the user: Anyone quoting "IPOs give X% average return" should be corrected — the spread of outcomes is enormous. Before applying to any IPO, treat the average as a weak guide and instead ask where that IPO's GMP and subscription profile falls relative to history.

Output 2 — Pearson Correlation Matrix

		Correlations				
		Log_retail	Log_HNI	Log_QIB	Net_sent	GMP_percent age
Log_retail	Pearson Correlation	1	.753**	.606**	.016	.671**
	Sig. (2-tailed)		.000	.000	.804	.000
	N	258	258	258	258	258
Log_HNI	Pearson Correlation	.753**	1	.733**	-.021	.673**
	Sig. (2-tailed)	.000		.000	.739	.000
	N	258	258	258	258	258
Log_QIB	Pearson Correlation	.606**	.733**	1	-.025	.651**
	Sig. (2-tailed)	.000	.000		.688	.000
	N	258	258	258	258	258
Net_sent	Pearson Correlation	.016	-.021	-.025	1	.036
	Sig. (2-tailed)	.804	.739	.688		.566
	N	258	258	258	258	258
GMP_percentage	Pearson Correlation	.671**	.673**	.651**	.036	1
	Sig. (2-tailed)	.000	.000	.000	.566	
	N	258	258	258	258	258

** . Correlation is significant at the 0.01 level (2-tailed).

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Variable: HNISubscription  Type: Number   Width:  12  Dec:  4
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Variable: GMPRs            Type: Number   Width:  12  Dec:  1
Variable: GMP_percentage   Type: Number   Width:  12  Dec:  4
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Actual SPSS output: correlation matrix across GMP, subscription, and sentiment

What this is: Pairwise correlation coefficients between GMP, log-transformed Retail/HNI/QIB subscription, and net sentiment.

Layman meaning: GMP moves closely with all three subscription categories (correlations of roughly 0.6 to 0.75) — when institutional and retail demand is high, GMP tends to be high too. Net sentiment shows almost no relationship with any other variable. This is the first clue that GMP and subscription may be measuring the same underlying thing, while sentiment measures something separate — and possibly irrelevant.

Actionable for the user: Before trusting multiple signals together, check whether they're actually independent. If two signals move together, using both doesn't double your information. This step justified running GMP-included and GMP-excluded models separately.

Output 3 — Regression Model 1: Full Model (GMP + Subscription + Sentiment)

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	GMP_percent age, Net_sent, Log_QIB, Log_retail, Log_HNI ^b	.	Enter

a. Dependent Variable: Listing_gain

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.799 ^a	.638	.631	24.32287

a. Predictors: (Constant), GMP_percentage, Net_sent, Log_QIB, Log_retail, Log_HNI

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	263059.204	5	52611.841	88.931	.000 ^b
	Residual	149083.718	252	591.602		
	Total	412142.922	257			

a. Dependent Variable: Listing_gain

b. Predictors: (Constant), GMP_percentage, Net_sent, Log_QIB, Log_retail, Log_HNI

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-8.729	3.073		-2.840	.005
	Log_retail	-3.196	3.844	-.051	-.831	.407
	Log_HNI	4.147	2.781	.102	1.491	.137
	Log_QIB	4.105	3.067	.079	1.338	.182
	Net_sent	.627	.941	.025	.667	.506
	GMP_percentage	1.141	.092	.705	12.367	.000

Actual SPSS output: Model Summary, ANOVA, and Coefficients — full model

What this is: The main regression — Listing_gain predicted using Log_retail, Log_HNI, Log_QIB, Net_sent, and GMP_percentage together ($R^2 = .638$, $F = 88.931$, $p < .001$).

Layman meaning: When all signals compete for influence in the same model, GMP overwhelmingly wins. Its effect (Beta = .705) is significant at $p < .001$, while every subscription variable and sentiment become statistically insignificant (p-values of .18 to .51). Once you know an IPO's GMP, checking subscription numbers barely changes your prediction of the listing gain.

Actionable for the user: If someone has limited time or data before an IPO closes, checking GMP alone gives them almost as much predictive information as checking everything. This is the single most decision-relevant output in the project.

Output 4 — Collinearity Diagnostics (Model 1)

Coefficients^a

Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	Log_retail	.383	2.609
	Log_HNI	.304	3.289
	Log_QIB	.416	2.401
	Net_sent	.993	1.007
	GMP_percentage	.442	2.264

a. Dependent Variable: Listing_gain

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions			
				(Constant)	Log_retail	Log_HNI	Log_QIB
1	1	4.265	1.000	.01	.01	.01	.01
	2	.986	2.080	.00	.00	.00	.00
	3	.372	3.388	.40	.02	.02	.01
	4	.178	4.893	.05	.18	.19	.00
	5	.129	5.759	.08	.63	.25	.20
	6	.071	7.738	.46	.17	.53	.78

Collinearity Diagnostics^a

Model	Dimension	Variance Proportions	
		Net_sent	GMP_percent age
1	1	.00	.01
	2	.98	.00
	3	.00	.17
	4	.01	.73
	5	.01	.01
	6	.00	.08

a. Dependent Variable: Listing_gain

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REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA COLLIN TOL
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT Listing_gain
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Actual SPSS output: Tolerance and VIF statistics

What this is: Tolerance and VIF (Variance Inflation Factor) statistics for each predictor in Model 1.

Layman meaning: All VIF values are below 3.3 — well under the common concern threshold of 5 or 10. Although GMP and subscription are correlated, they aren't so overlapping that the regression results become unstable or untrustworthy.

Actionable for the user: This is a technical checkpoint that backs the business conclusion: the "GMP dominates" finding is a real statistical result, not a side-effect of overlapping variables confusing the model.

Output 5 — Regression Model 2: Subscription-Only (GMP Removed)

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Net_sent, Log_retail, Log_QIB, Log_HNI ^b	.	Enter

a. Dependent Variable: Listing_gain

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.647 ^a	.419	.410	30.77202

a. Predictors: (Constant), Net_sent, Log_retail, Log_QIB, Log_HNI

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	172572.873	4	43143.218	45.562	.000 ^b
	Residual	239570.049	253	946.917		
	Total	412142.922	257			

a. Dependent Variable: Listing_gain

b. Predictors: (Constant), Net_sent, Log_retail, Log_QIB, Log_HNI

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics
		B	Std. Error	Beta			Tolerance
1	(Constant)	-15.041	3.834		-3.923	.000	
	Log_retail	11.826	4.614	.188	2.563	.011	.426
	Log_HNI	9.745	3.472	.241	2.807	.005	.312
	Log_QIB	15.261	3.709	.292	4.115	.000	.456
	Net_sent	1.366	1.188	.055	1.149	.252	.997

Actual SPSS output: Model Summary, ANOVA, and Coefficients — GMP excluded

What this is: The same regression re-run with GMP deliberately excluded, leaving only Log_retail, Log_HNI, Log_QIB, and Net_sent as predictors ($R^2 = .419$).

Layman meaning: With GMP out of the picture, subscription demand becomes clearly important — Retail ($p=.011$), HNI ($p=.005$), and QIB ($p<.001$) subscription are all statistically significant, together explaining about 42% of the variation in listing gains. Subscription data genuinely carries predictive value — it only looked unimportant earlier because GMP was already capturing the same information.

Actionable for the user: If GMP data is ever unavailable, unreliable, or suspected of manipulation (a real risk, since it's an informal, unregulated figure), subscription numbers are a legitimate fallback signal — not a weak substitute.

Output 6 — Regression Model 3: Raw Subscription Values + GMP + Sentiment

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percent age ^b	.	Enter

a. Dependent Variable: Listing_gain

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.811 ^a	.658	.652	23.63337

a. Predictors: (Constant), QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percentage

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	271391.779	5	54278.356	97.180	.000 ^b
	Residual	140751.143	252	558.536		
	Total	412142.922	257			

a. Dependent Variable: Listing_gain

b. Predictors: (Constant), QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percentage

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-5.358	2.042		-2.624	.009
	Net_sent	.499	.915	.020	.545	.586
	GMP_percentage	.935	.098	.578	9.502	.000
	Retail Subscription	.088	.089	.055	.994	.321
	HNI Subscription	.046	.017	.161	2.721	.007
	QIB Subscription	.067	.033	.108	2.034	.043

Actual SPSS output: Model Summary, ANOVA, and Coefficients — raw subscription values

What this is: A third specification using untransformed (raw, not log) subscription multiples alongside GMP and sentiment ($R^2 = .658$, unfiltered sample).

Layman meaning: Even using raw subscription figures, GMP remains dominant — but HNI Subscription ($p=.007$) and QIB Subscription ($p=.043$) now retain a small but genuine independent effect alongside GMP. Institutional demand isn't completely absorbed by GMP — a bit of unique signal remains.

Actionable for the user: For a more advanced screening rule: GMP first, but if HNI or QIB subscription is unusually weak despite strong GMP, treat that as a secondary risk flag rather than ignoring it entirely.

Output 7 — Outlier-Trimmed Regression (N = 239)

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percent age ^b	.	Enter

a. Dependent Variable: Listing_gain

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.766 ^a	.586	.577	22.90473

a. Predictors: (Constant), QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percentage

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	173091.086	5	34618.217	65.986	.000 ^b
	Residual	122238.012	233	524.627		
	Total	295329.098	238			

a. Dependent Variable: Listing_gain

b. Predictors: (Constant), QIB Subscription, Net_sent, Retail Subscription, HNI Subscription, GMP_percentage

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-4.770	2.050		-2.327	.021
	Net_sent	.679	1.119	.026	.607	.545
	GMP_percentage	.934	.114	.588	8.183	.000
	Retail Subscription	.057	.116	.027	.493	.622
	HNI Subscription	.047	.024	.112	1.942	.053
	QIB Subscription	.068	.038	.113	1.820	.070

Actual SPSS output: Model Summary, ANOVA, and Coefficients — outlier-trimmed sample

What this is: After computing Z-scores and identifying extreme cases, the sample was reduced from 258 to 239 IPOs, and the regression was re-run to check whether earlier findings depended on a few unusual listings.

Layman meaning: R^2 drops somewhat (to .586) once extreme cases are removed — expected, since some explanatory power came from a handful of dramatic winners or losers. But GMP is still the dominant, statistically significant predictor after this trim. The core conclusion doesn't change; it just becomes more conservative and trustworthy.

Actionable for the user: This is the proof that "trust GMP first" isn't a fluke caused by one or two spectacular IPO listings — it holds for the typical, non-extreme IPO too.

Business Takeaway

When multiple signals overlap, testing them together — not in isolation — reveals which one actually carries information. Here, that signal is GMP, with subscription demand as a credible secondary check and media sentiment carrying no measurable value.